SOFC DIGITAL TWIN · PART LOAD

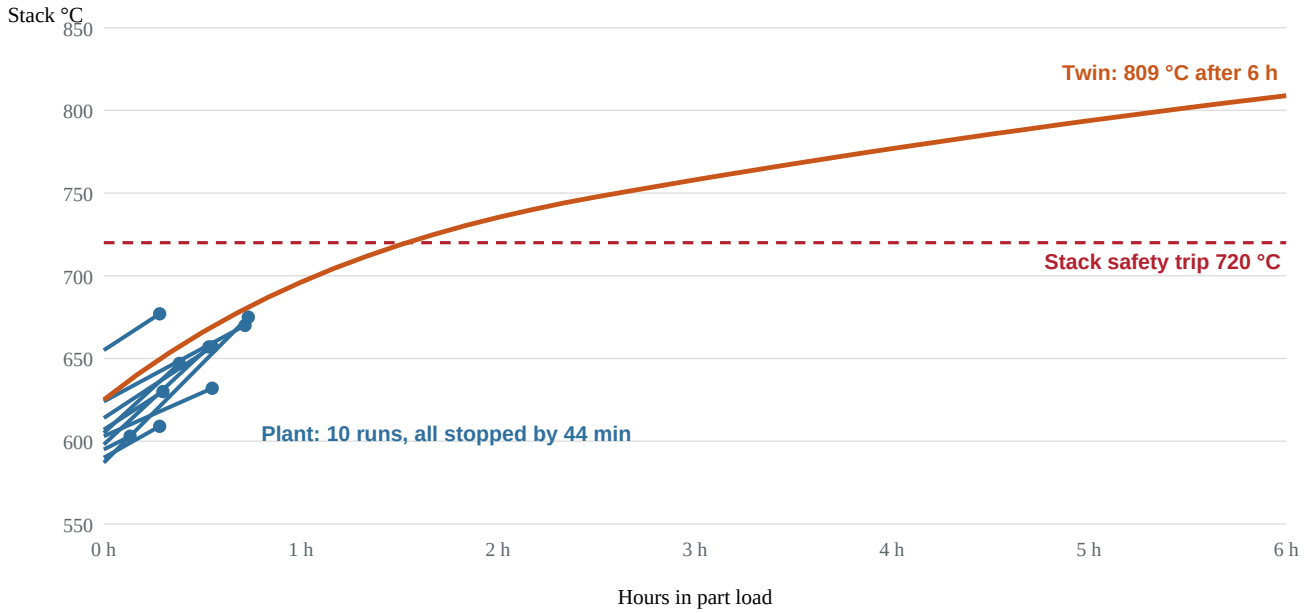
Why does the twin's stack keep getting hotter?

Contents: 1 The answer · 2 Why it climbs · 3 How to stop it · 4 Twin vs plant in part load · 5 What changed

Because in part load the twin keeps adding about 100 °C per hour of heat, and nothing in the twin turns the heat down or the cooling up when the stack gets hot.

Fixed on 2026-11-04: the twin matches the plant part-load runs to about 4 °C. It now holds the air-in like the plant, trips at 720 °C and has a stack keeper (sections 4 and 5).

The twin was taught from plant data. But the plant never stayed in part load for more than 44 minutes. So the twin has never seen what the plant does to hold the stack steady. Left alone in part load, the twin keeps warming until it settles near 920 °C, far above the 720 °C safety trip.



Orange: the twin started at the plant's usual part-load settings (stack 625 °C, air 9000 ln/min, natural gas 137 ln/min, 6.4 A per string). Blue: the 10 real part-load runs in the 39 days of plant data. Every real run was stopped within 44 minutes.

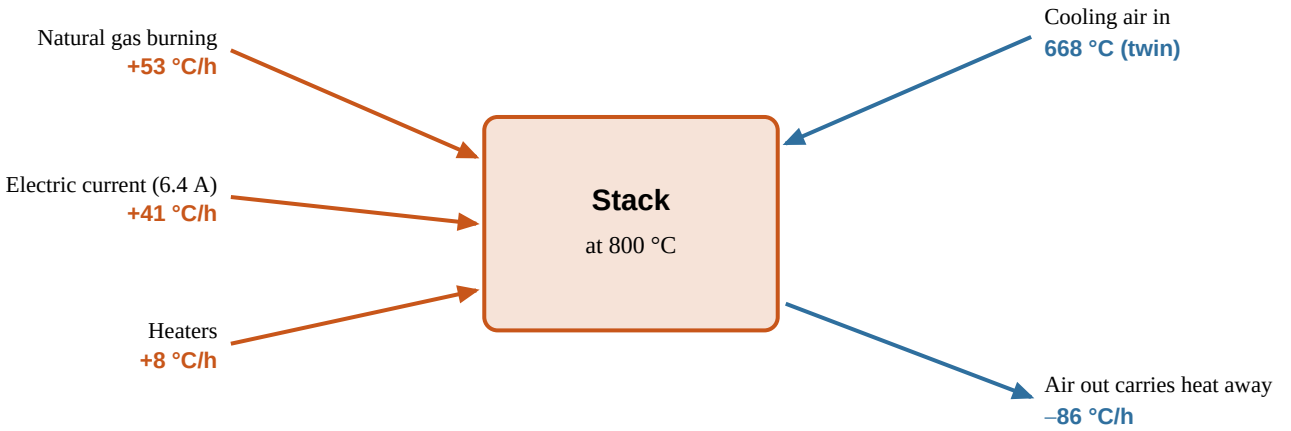
What	Twin	Real plant
Warming speed at the start of part load	+94 °C/h	+77 °C/h (median)
Time in part load	as long as you let it	8–44 min
Hottest stack seen	921 °C after 24 h	677 °C
Crosses the 720 °C trip	after 1.6 h	never
Safety lamp on the twin screen	stays green	–

If you remember one thing: the twin is a car with the accelerator held down and no driver. The plant data never shows the driver for longer than 44 minutes.

Why it climbs

Think of the stack as a pot on a stove. Three burners add heat. The cooling air flowing through it is the only thing taking heat away. The pot stops warming only when the two are equal.

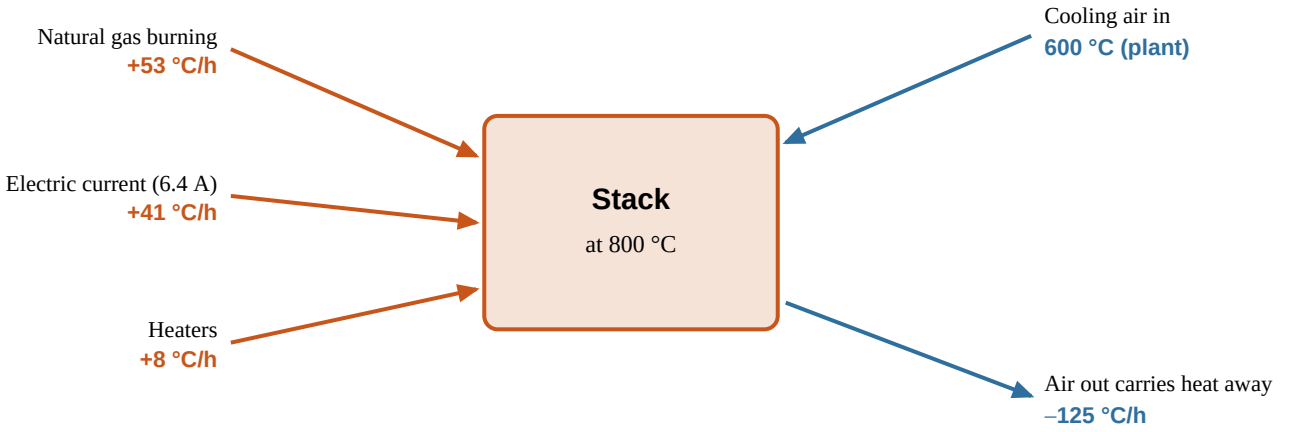
Twin's cooling air (follows the stack):



Heat in: about +101 °C/h, same at any temperature

Net: still warming, +15 °C/h

Plant's cooling air (held near 600 °C):



Heat in: about +101 °C/h, same at any temperature

Net: cooling, –24 °C/h

Heat balance with the stack at 800 °C, part-load settings, air 9000 ln/min, heaters on. Numbers come from the twin's own formulas; 'air out' includes the small heat loss through the insulation.

1 The heat coming in is fixed. Gas burning, current and heaters give about +101 °C/h whatever the stack temperature is. The twin learned these numbers from the first half hour of part load, when the plant was also warming fast (+77 °C/h). They tell how fast the stack rises, not where it stops.

2 The cooling air gets hotter as the stack gets hotter. In the twin, the air going into the stack is always 83 % of the way to the stack temperature. In the plant, the air-in temperature (sensor TIRC 2.06) stays near 600 °C in part load (mostly 580–610 °C).

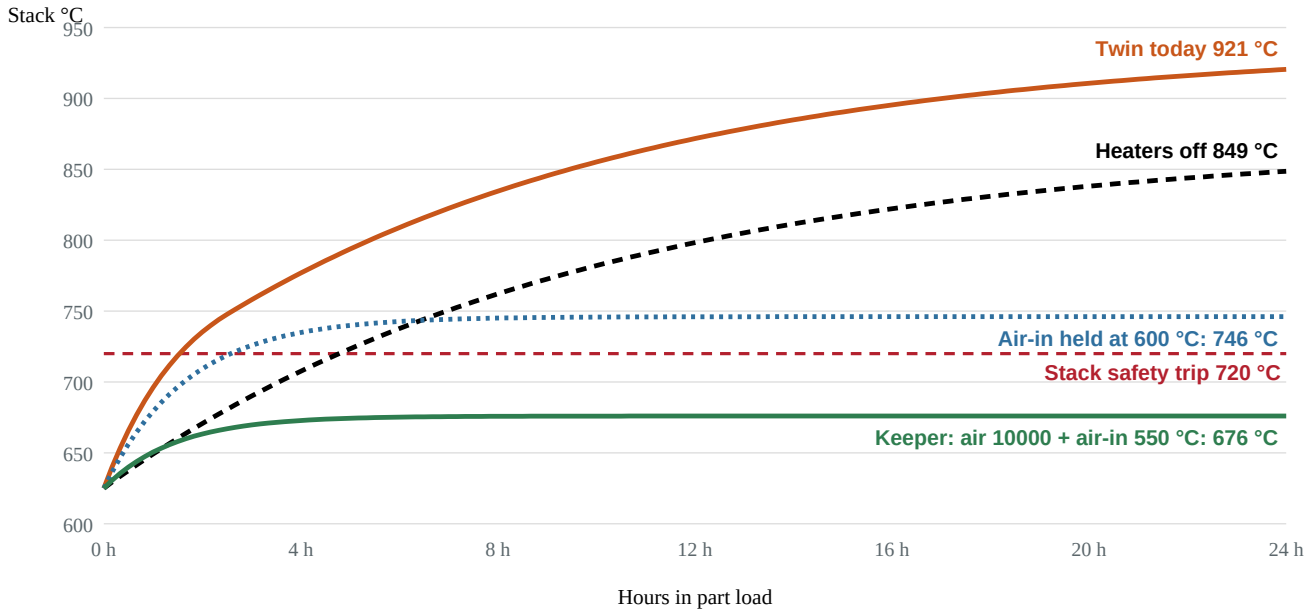
3 Nobody turns a knob. The twin has no temperature keeper. In the plant every part-load run ended within 44 minutes: 3 to air cooling, 2 to hot standby, 2 back to fuel switch, 1 to shutdown, 2 by a safety trip.

4 The safety lamp cannot see it. The 'Stack safety' lamp reads a look-up sensor capped at 689 °C (highest seen in plant part load). It never reaches the 690 °C warning, so it stays green even at 900 °C.

If you remember one thing: the twin's cooling air warms up together with the stack, and nothing in the twin ever turns the heat down.

How to stop it

The twin needs a 'driver': a temperature keeper that turns the cooling up when the stack gets hot, the way the plant's control system and operators do.



Quick what-if runs on the twin's formulas from the same part-load start (625 °C), not yet checked against plant data.

Version	After 1 h (°C)	After 24 h (°C)	Crosses 720 °C
Twin today (heaters on)	696	920	after 1.6 h
Heaters off	649	849	after 4.8 h
Heaters off, air-in held at 600 °C (as in the plant)	679	746	after 2.6 h
… plus air raised to 10000 ln/min	675	725	after 4.3 h
… plus air-in lowered to 550 °C	651	676	never

What I suggest, in order

1 Make the safety lamp read the twin stack (warn 690 °C, trip 720 °C). 2 Hold the cooling air at the plant's 600 °C in part load and fuel switch. 3 Add a stack temperature keeper: above a limit, raise air towards 10000 ln/min and lower the air-in target (test: settles at 676 °C). 4 Let the twin trip at 720 °C, as the plant does.

Decisions for you

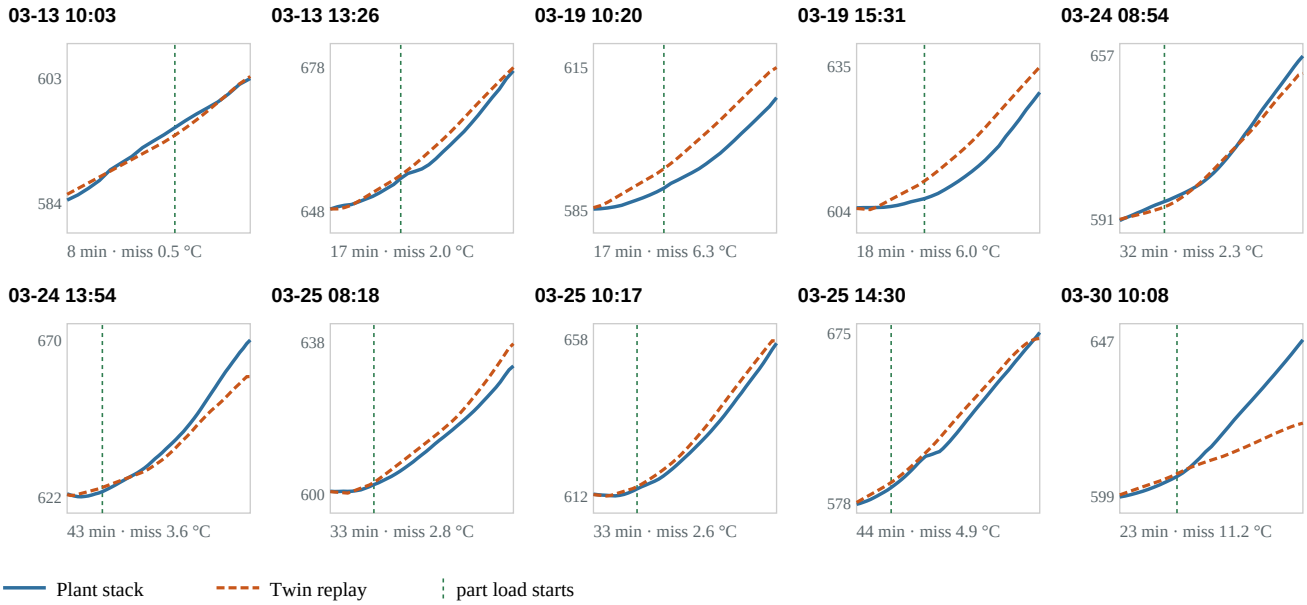
	Question	My suggestion
A	What stack temperature should the keeper hold in part load?	About 680 °C (plant runs ended at up to 677 °C)
B	Which knob should the keeper turn: air flow, air-in temperature, current or gas?	Air flow first, then air-in temperature
C	Should the twin trip itself at 720 °C?	Yes, as the plant does
D	Can we get plant data with a part load longer than 1 hour?	Yes if possible: the only way to check where the real stack settles

Where in the twin: twin_2d/app.js, functions rate and tinTarget, and the safety lamp in render.

If you remember one thing: give the twin a temperature keeper and fix the safety lamp. The plant has never shown where it settles in part load.

Twin vs plant in part load

Given the plant's own knob settings minute by minute, the twin follows the real stack in all 10 part-load runs. The average miss is 4.2 °C. So the heat balance is right where the plant has been; the trouble only starts after the 44 minutes the plant never went past.



Each small chart is one part-load run from the 39 days, starting 10 minutes before part load. Blue: measured stack. Orange dashed: the twin given the same air, gas, current, heater and air-heater target. Each chart has its own scale.

Run	Length	Plant °C/h	Twin °C/h	Average miss °C	Miss at end °C
2026-03-13 10:03	8 min	+64	+77	0.5	+0
2026-03-13 13:26	17 min	+84	+84	2.0	+1
2026-03-19 10:20	17 min	+71	+79	6.3	+6
2026-03-19 15:31	18 min	+80	+86	6.0	+5
2026-03-24 08:54	32 min	+113	+104	2.3	-7
2026-03-24 13:54	43 min	+66	+48	3.6	-11
2026-03-25 08:18	33 min	+56	+65	2.8	+6
2026-03-25 10:17	33 min	+80	+80	2.6	+1
2026-03-25 14:30	44 min	+123	+114	4.9	-3
2026-03-30 10:08	23 min	+116	+43	11.2	-26

The largest miss (30 March, 26 °C at the end) is a run with very low current (2.8 A) where the plant still warmed fast; one run is too few to change the twin for.

What the plant does in part load

Measured	Plant	Twin before
Air heater (EAPH)	stays on, 8.6 kW median	on after a stage jump; the playbook left its target at 500 °C
Air heater target	580 °C fuel switch, 600 °C part load	595 / 625 °C (jump), 500 °C (playbook)

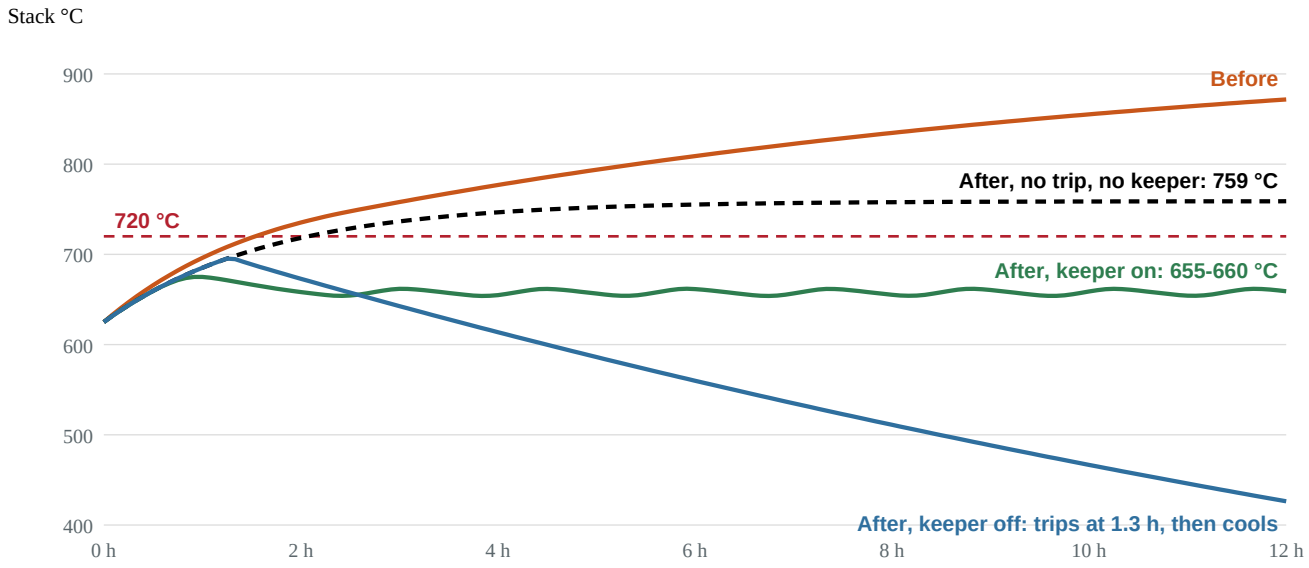
Air going into the stack	held at the target, even when the heat recovery runs hotter (median 0 K off)	followed the stack once the heat recovery got hotter than the target
Longest part load	44 min, stack up to 677 °C	unlimited

If you remember one thing: inside the plant's 44 minutes the twin is right to within about 4 °C. It overheated only because it was allowed to run on where the plant data never went.

What changed

The twin now copies what the plant does in part load, stops itself at the plant's safety limit, and has an optional keeper that holds the stack below that limit.

	Change	Why (plant data)
1	Air-in held at the air-heater target while the heater is on (the 3-way valve TWV-CA mixes in cooler bypass air)	Plant: air-in minus target median 0 K with the heater on, also when the heat recovery was about 24 K hotter
2	'Go to part load' and the stage buttons use the plant's targets with heaters on: 580 °C fuel switch, 600 °C part load	Plant medians; air heater on at 8.6 kW in part load
3	Safety lamp follows the twin stack; at 720 °C on the stack safety thermocouple the twin trips (20 min purge, then air-flow cooling)	Plant trip limit 720 °C; that thermocouple reads about 24 °C above the stack average
4	Stack keeper switch (on by default, twin only): above 660 °C raise air towards 10000 ln/min, then lower the air-heater target (not below 500 °C); hand both back under 655 °C	Not in the plant data: every plant part load was ended within 44 min



Hours after entering part load (twin runs, not plant data)

Check	Before	After
10 plant part-load runs, average miss	4.2 °C	4.2 °C
8 earlier test stretches, average miss	12.7 °C	12.7 °C
Left in part load, keeper off	872 °C after 12 h, no warning	trips after 1.3 h (stack 697 °C)
Left in part load, keeper on	–	peak 675 °C, then 655-661 °C

Still open for you

	Question	What I set for now

A	Keeper limit	660 °C stack (safety thermocouple stays at or below 697 °C)
B	Keeper knobs	Air first, then air-heater target
C	Keeper on by default?	Yes; switch it off to see the plant-like trip
D	A longer part-load test on the plant	Needed to check where the real stack settles

Files: twin_2d/app.js, twin_2d/index.html, twin_2d/README.md.

If you remember one thing: the twin matched the plant all along; now it also stops where the plant would stop.